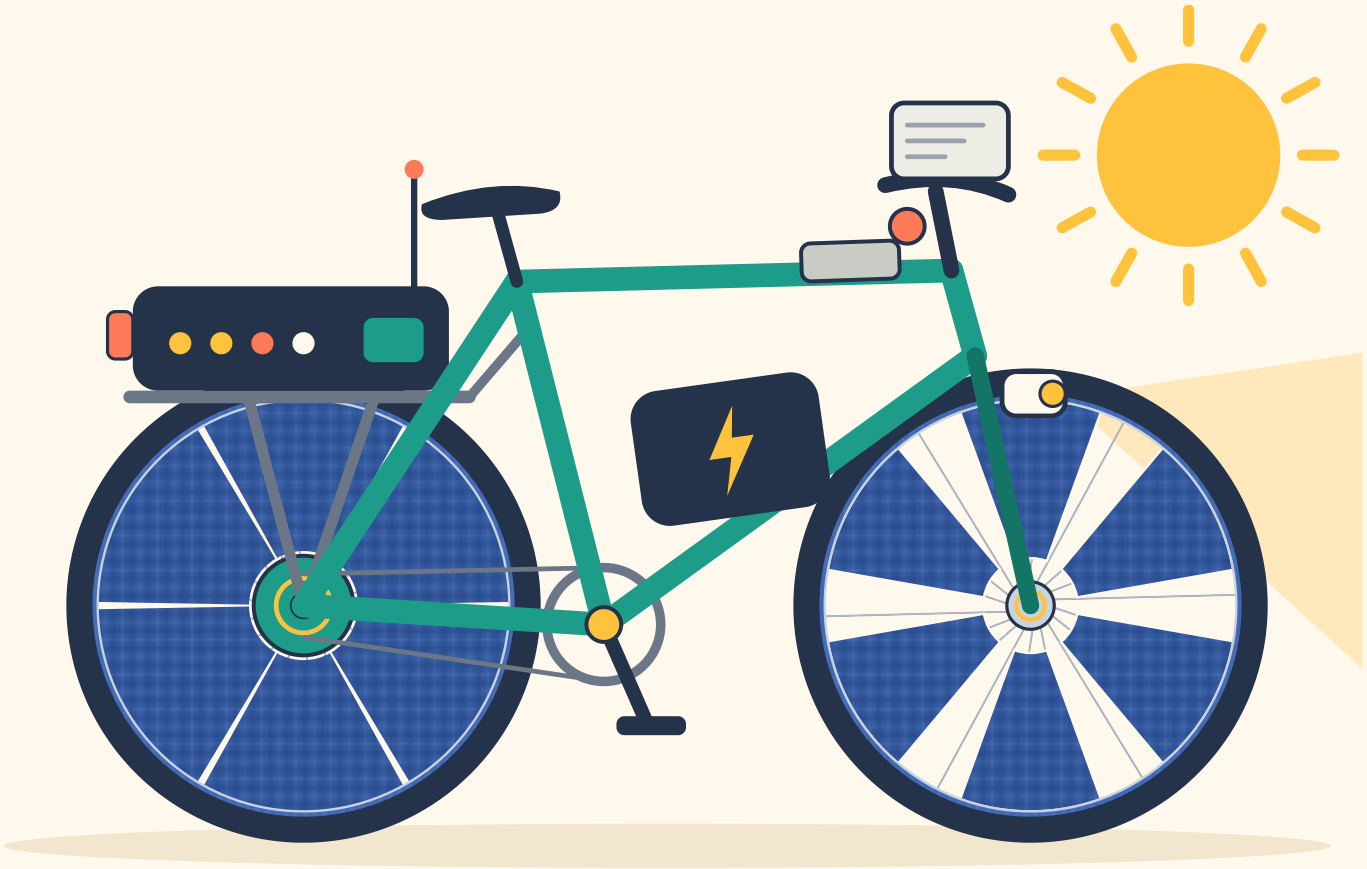
THE FRIENDLY GUIDE · CONCEPT V0.3

Pedal Power Station

An e-bike that never plugs into the wall. Your legs and the sun are the whole charger, and the wheels themselves are solar panels.



Solar wheels · Assist to 15 mph · No wall plug · Auto headlight · USB-C + Lightning · Off-grid messaging

540 Wh
battery, filled only by you and the sun

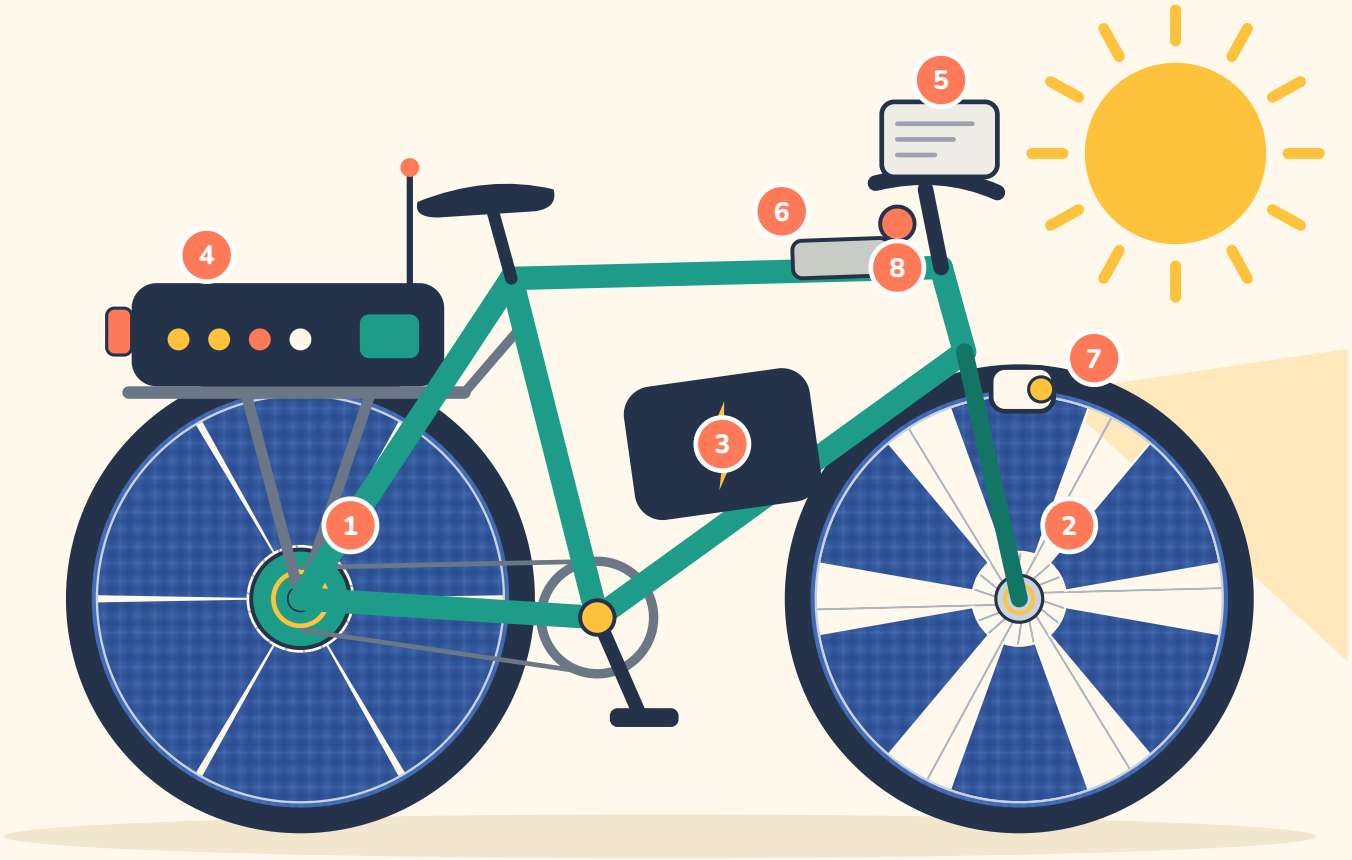
~7 mi
of assist per hour of steady pedaling

~105 W
of wheel solar when parked with the Sun Wings up

MEET YOUR BIKE

Everything has a job

Ride it around the neighborhood, park it on its stand as a home generator, or just leave it in the sun. Every number below points to a part on the bike.

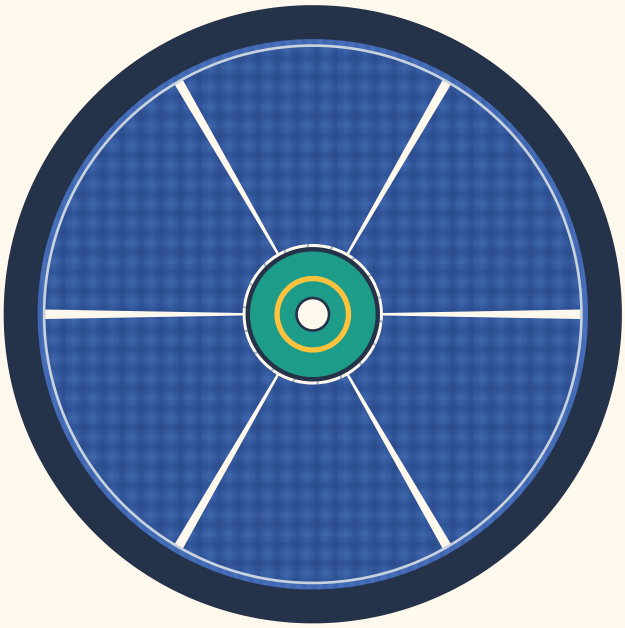


1. **Solar disc + hub motor.** The rear wheel is a full solar disc. Its hub pushes you along and works as a generator.
2. **Solar petals.** Six solar petals in the front wheel, with gaps so side winds pass through.
3. **36V battery.** Sits low in the frame. It has no charging port at all.
4. **Power trunk.** USB-C and USB-A ports, a 13.8V radio/car port, and the brains.
5. **Big screen.** E-paper Raspberry Pi display for messages, maps and energy history.
6. **Status screen.** Always on: speed, battery, time, and miles of assist left.
7. **Headlight.** Turns on by itself at dusk, whether you are riding or on the stand.
8. **Cord pod.** Pull-out USB-C and Lightning cords for phones and tablets.

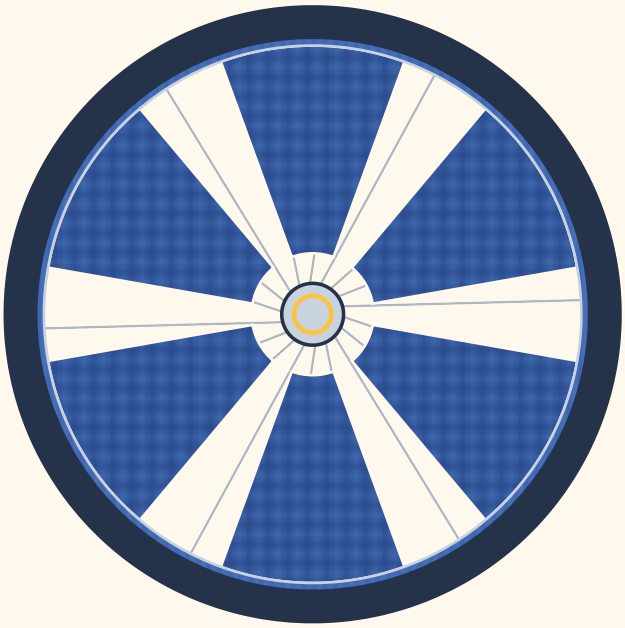
THE STAR FEATURE

Sun Wheels

Solar cells are built into the spoke area and the rims, on both sides of both wheels. Park it facing the sun and the wheels charge the battery all day.



Rear: full solar disc



Front: six solar petals

How it works

1. **Cells on both faces.** Thin, flexible solar cells sit under a tough clear cover that snaps onto the spokes.
2. **Rim strips.** A narrow band of cells wraps each rim sidewall for a little extra.
3. **Six slices.** Each wheel face is split into six slices, so the frame's shadow dims only one slice, not the whole wheel.
4. **Spinning joint.** A small sealed slip ring at each hub carries the power from the turning wheel to the battery.

Why wheel solar works. A wheel spins flat, in its own plane, so its angle to the sun stays the same as it turns. Whichever side faces the sun does the work.

~80 W
cells facing the sun

50–150 Wh
clear day, wings down

5–15 mi
of assist from one sunny day

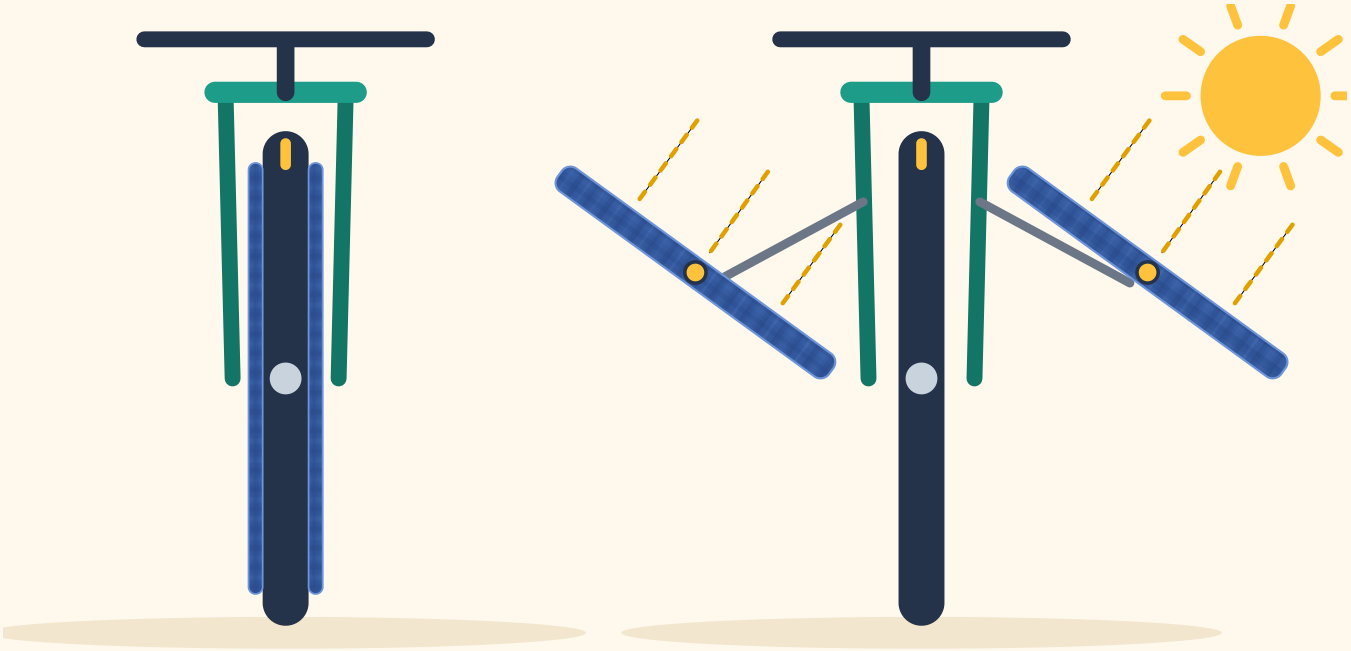
~3 kg
added to the wheels

Good to know. A solid disc catches side winds, so only the rear is a full disc; the front uses petals with open gaps. The best results come from parking with one wheel face toward the sun. The Sun Finder on the big screen tells you which way to turn the bike. When parked, the front panels can also lift up and tilt toward the sun (next page).

PARKED? OPEN THE WINGS

Sun Wings

A flat wheel only gets full sun on one side. So when you park, the front wheel's two solar panels lift up through a slit and tilt toward the sun, and **both** of them get direct sunlight.



Riding: panels locked flat

Parked: lifted and tilted to the sun

How to open them

1. **Stop with the yellow mark on top.** It lines up the slit in the wheel cover with the fork.
2. **Lift each panel up through the slit.** It slides out of the wheel and clicks onto a swing arm on the fork.
3. **Tilt both toward the sun.** The Sun Finder on the big screen shows the best angle for the time of day.
4. **Slide them home before you ride.** A sensor in each latch keeps the motor off until both panels are locked back in the wheel.

2x
front-wheel solar when parked

~105 W
whole bike, wings up

~250 Wh
best case, clear summer day

~25 mi
of assist from that

Why it helps. Riding, only one side of the wheel faces the sun and it stands straight up. With the wings up, both panels get sunlight and they lean toward it, which collects more than standing straight.

Good to know. The wings are for parking only, and the bike will not ride with them up. Fold them down in strong wind. Winter days are shorter, so expect about half the summer number.

HOW YOU USE IT

Three ways to ride, one way to rest

Pick a setting on the status screen. Switching between earning energy and spending it is one button press.

Harvest

A light pull, like riding into a gentle breeze. Your effort charges the battery.

Great on flats and downhills

Neutral

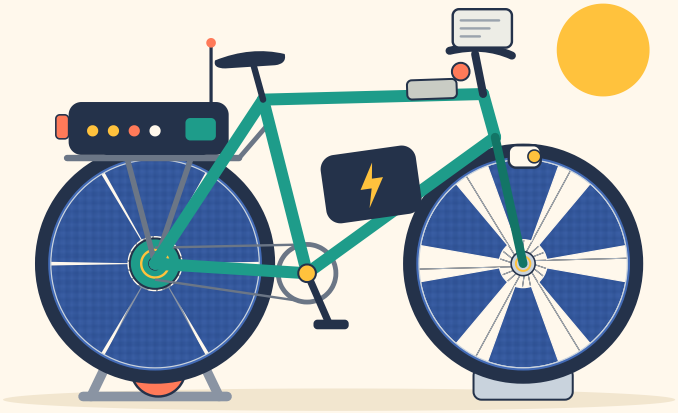
No help, no drag. Just a (slightly heavy) bicycle.

For easy cruising

Assist

A friendly push while you pedal, up to 15 mph. No throttle.

Great for hills and headwinds



Station mode

Flip down the stand, rest the front wheel on the block, and pedal. A flywheel gives a smooth, road-like feel while the hub banks every watt.

The headlight doubles as a work light if you are pedaling during a blackout.

Built-in good habits

15 mph cap

Assist fades out between 14 and 15 mph. You can still pedal faster on your own.

25% kept in reserve

Assist stops at 25% so there is always power left for lights, phones and radios.

No wall plug

There is no charging port. Every bit of assist is energy you or the sun made.

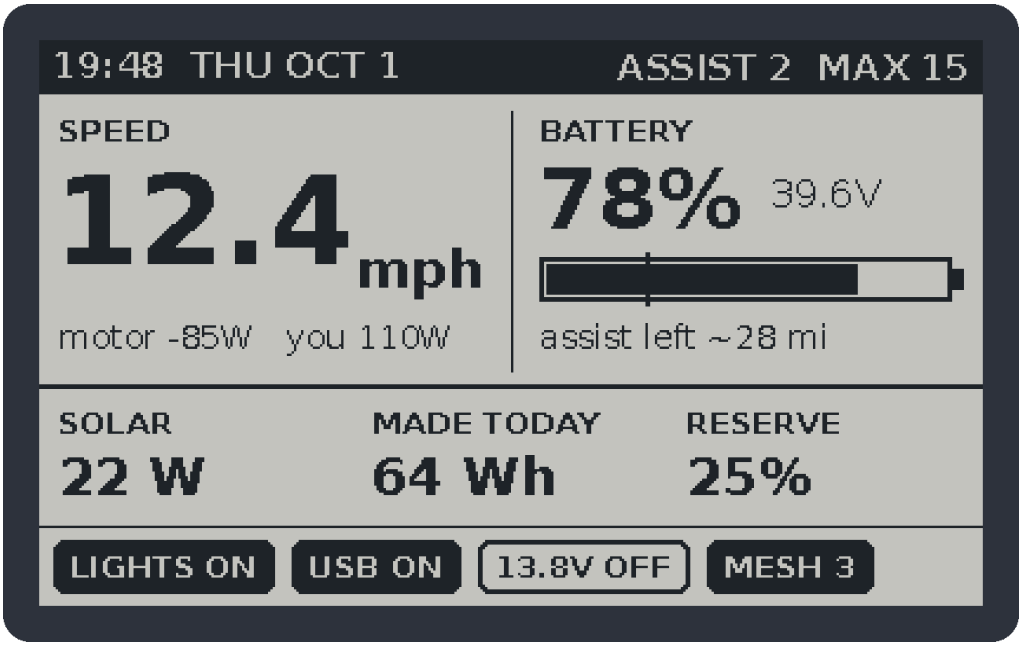
Class 1 e-bike

Pedal-assist only, 500W, 15 mph: within Michigan Class 1 rules. Check local trails.

YOUR DASHBOARD

Two screens, no glare

Neither screen has a backlight. They get easier to read in bright sun, and sip almost no power. Tiny front lights come on at night.

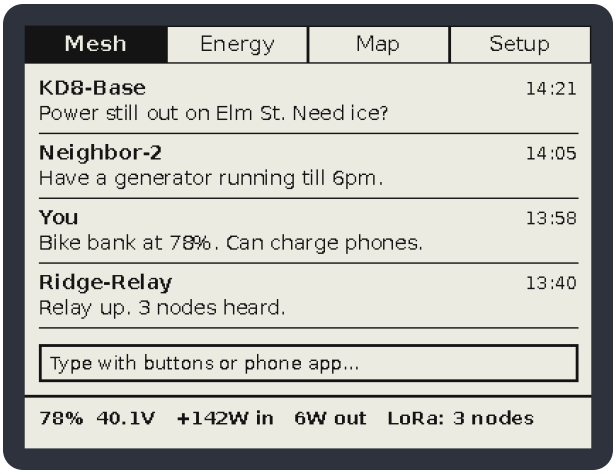


Status screen

Lives on the top tube and is always on. Shows the time, your speed (or pedal watts on the stand), battery, miles of assist left, and how much the sun is adding right now.

Run by a tiny Pico

A Raspberry Pi Pico runs this screen and the safety rules, using about a tenth of a watt. It works even when the big computer is asleep.



The big screen

A 4.2-inch e-paper display on the handlebars, driven by a Raspberry Pi Zero 2 W that wakes when you need it.

- **Mesh:** text neighbors with no cell service
- **Energy:** what you and the sun made
- **Sun Finder:** which way to park the bike
- **Map:** offline neighborhood map

YOUR EFFORT, PUT TO USE

What one hour of pedaling buys

A steady, comfortable 100 watts for an hour banks about 72 Wh. Here is what that gets you.

~7 miles
of e-assist at up to 15 mph

~5 phones
charged from empty to full

~12 hours
of headlight and tail light

~60 hours
of screens and mesh messaging

Ready when the power goes out

The bike keeps a quarter of its battery for exactly this.

Charge anything
Pull-out USB-C and Lightning cords, plus fast-charge USB-C and USB-A ports.

Power a radio
A steady 13.8V port runs ham and GMRS radios, the standard they expect.

Top up a car battery
A clamp lead with reverse-polarity protection helps a weak battery recover.

Text with no towers
A LoRa mesh radio passes messages neighbor to neighbor.

Light the way
The headlight comes on at dusk, riding or on the stand.

Keep charging
Park it, raise the Sun Wings, and the wheels keep topping it up.

THE QUICK SPECS

At a glance

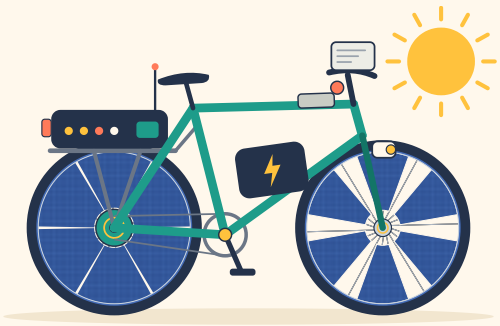
Battery	36V LiFePO4, 15Ah (about 540 Wh), built-in protection, no wall-charge port
Charged by	Pedaling (riding or on the stand), braking, and the solar wheels
Solar wheels	Rear full disc + front six petals + rim strips, both faces. Front side panels lift up and tilt toward the sun when parked. About 80 W riding, about 105 W parked with wings up
Motor	500W gearless rear hub, pedal-assist only, 15 mph cap; also the generator
Lights	Auto-on headlight and tail/brake light
Device charging	Pull-out USB-C and Lightning cords; USB-C fast charge and USB-A ports
Radio and car	Steady 13.8V 10A port, car-battery clamp lead
Screens	2.7in always-on status screen + 4.2in e-paper big screen, no backlights
Brains	Raspberry Pi Pico 2 (always on) + Raspberry Pi Zero 2 W (wakes on demand)
Messaging	Meshtastic LoRa mesh radio for off-grid texting
Added weight	About 18-20 kg over the starting bike
Estimated build cost	About $1,500-2,750 including a used starter bike

Keeping it real

This is a concept, not a finished engineered design. Before regular riding, have someone who knows e-bike and lithium battery wiring look it over. The brakes must cut the motor instantly, every time.

What's next

Start with a bench test: the hub, the battery and one solar wheel on a stand. Then add the screens, the stand, and finally ride mode. The full build plan is in the technical spec.



Your legs and the sun. That's the whole charger.